RUBEL ROY

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Professional Summary

Data Science master's student with hands-on experience in Python, statistics, and machine learning. Demonstrates strong ability to learn new concepts, explore complex datasets, and transform data into actionable insights. Recognized for curiosity, analytical problem-solving, and effective collaboration within teams. Driven to continuously develop skills and contribute to impactful AI and data science initiatives.

Education

St. Xavier's College — Kolkata, West Bengal,India
Master of Science in Data Science — *Graduation Date [2027]*

Gurudas College — Kolkata, West Bengal,India
Bachelor of Science in Physics — *Graduation Date [2024]*

Technical Skills

Programming: Python, SQL, R
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM
Data Analysis: Pandas, NumPy, SciPy
Data Visualization: Matplotlib, Seaborn, Tableau
Deployment & APIs: FastAPI, Flask, Streamlit
Tools & Environment: Git, GitHub, Linux, Jupyter Notebook, VS Code, Excel

Projects

Healthcare ML Platform — End-to-End Medical ML System — 2026 - Ongoing
Python | Scikit-learn | XGBoost | LightGBM | Streamlit | Pandas

- Built a modular ML platform on two real hospital datasets (Pima Indians — 768 records, Diabetes 130-US — 101,766+ records) implementing three core ML systems: **Regression** (patient readmission risk score 0–100%), **Classification** (diabetic vs non-diabetic prediction), and **Recommendation** (treatment suggestions via kNN) — alongside Clustering and Anomaly Detection.
- Performed deep EDA across two heterogeneous datasets — handled biologically impossible zeros as hidden missing values (48.7% of Insulin column), "?" string nulls in 130-US dataset, age bracket-to-numeric conversion, and class imbalance using median imputation and mode filling.
- Engineered 4 new clinical features (BMI category, Age group, Glucose level, composite Risk score) using WHO and ADA medical thresholds; applied StandardScaler and LabelEncoder with fitted objects saved as `.pkl` files for consistent inference — confirmed 65/35 stratified train/test split.
- Built a kNN recommendation engine trained on 101,766+ hospital admissions — finds 5 most similar past patients by Euclidean distance across 6 clinical features and returns ranked medication recommendations (insulin, metformin, rosiglitazone) with per-medication confidence scores.

Fitbit Fitness Data Analysis — March 2025
Python | Pandas | Matplotlib | Seaborn

- Performed exploratory data analysis on 900+ Fitbit activity records to study user behavior patterns.
- Identified correlation between total steps and calorie expenditure using statistical analysis.
- Investigated relationship between sleep duration and time spent in bed.
- Created visualizations to highlight daily activity and health trends.

Certifications

Supervised Learning in Python –DataCamp	*Feb 2026*
Unsupervised Learning in Python –DataCamp	*Feb 2026*
Deep Learning for Images with PyTorch – DataCamp	*Feb 2026*
Django for Everybody Specialization – University of Michigan / Coursera	*Mar 2025*
Google Data Analytics Professional Certificate – Coursera	*Dec 2024*